

CLINICAL TRIALS AND OBSERVATIONS

CME Article

First-line ibrutinib plus venetoclax for nonblastoid mantle cell lymphoma in patients ≥ 65 years old or with *TP53* mutations

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KEY POINTS

- First-line ibrutinib plus venetoclax showed promising efficacy with a high CR rate and durable remissions in patients with nonblastoid MCL.
- Ibrutinib plus venetoclax may be an option for patients with previously untreated nonblastoid MCL aged ≥ 65 years or of any age with a *TP53*m.

The phase 3 SYMPATICO study included an open-label cohort to evaluate first-line ibrutinib plus venetoclax in patients with nonblastoid mantle cell lymphoma (MCL) aged ≥ 65 years ($n = 65$) or ≥ 18 years with a *TP53* mutation (*TP53*m; $n = 11$). Eligible patients received oral ibrutinib 560 mg once daily and venetoclax (5-week ramp-up to 400 mg once daily) for 2 years, then single-agent ibrutinib 560 mg until disease progression or unacceptable toxicity. In total, 78 patients were enrolled. With a median time on study of 40.5 months, the complete response (CR) rate was 69% (95% confidence interval [CI], 58-79), and the overall response rate was 95% (95% CI, 87-99). The median duration of response was 37.1 months (95% CI, 30.3 to not estimable [NE]). The median progression-free survival (mPFS) was 40.2 months (95% CI, 29.4 to NE); median overall survival (OS) was not reached (3-year OS rate, 79% [95% CI, 68-86]). In patients aged ≥ 65 years, CR rates were 76% (no *TP53*m) and 44% (with *TP53*m), mPFS was 40.2 and 22.0 months, and 3-year OS was 85% and 66%, respectively. In adult patients aged < 65 years (with *TP53*m), the CR rate was 73%, mPFS was 15.4 months, and 3-year OS was 73%. The most common treatment-emergent adverse events were diarrhea (49%), fatigue (37%), neutropenia (35%), and COVID-19 (32%). First-

line ibrutinib plus venetoclax showed promising efficacy, with high CR rates and durable remissions, in patients with previously untreated nonblastoid MCL and may be an option for patients aged ≥ 65 years or patients of any age with *TP53*m. The trial was registered at www.clinicaltrials.gov as NCT03112174.



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Disclosures

CME questions author Laurie Barclay, freelance writer and reviewer, Medscape Education, declares no competing financial interests.

Learning objectives

Upon completion of this activity, participants will:

1. Describe the efficacy of first-line ibrutinib plus venetoclax in patients aged ≥ 65 years ($n = 65$) with mantle cell lymphoma (MCL), or aged ≥ 18 to < 65 years with MCL with a *TP53* mutation ($n = 11$), based on findings from an open-label cohort in the phase 3 SYMPATICO study
2. Determine the safety of first-line ibrutinib plus venetoclax in patients aged ≥ 65 years ($n = 65$) with MCL or in patients aged ≥ 18 to < 65 years with MCL with a *TP53* mutation ($n = 11$), based on findings from an open-label cohort in the phase 3 SYMPATICO study
3. Identify clinical implications of the efficacy and safety of first-line ibrutinib plus venetoclax in patients aged ≥ 65 years with MCL or aged ≥ 18 to < 65 years with MCL with a *TP53* mutation, based on findings from an open-label cohort in the phase 3 SYMPATICO study

Release date: September 17, 2026; Expiration date: September 17, 2027

Introduction

Mantle cell lymphoma (MCL) is a subtype of B-cell lymphoma that often has an aggressive clinical course and a poor prognosis.¹ In young, fit patients with MCL, first-line treatment with intensive chemoimmunotherapy followed by autologous hematopoietic stem cell transplant and rituximab maintenance provides high response rates and progression-free survival (PFS) of up to 9 years.²⁻⁴ In addition, the TRIANGLE study has shown that intensive chemoimmunotherapy in combination with ibrutinib without stem cell transplant may be an acceptable alternative.⁵ However, intensive chemoimmunotherapy is associated with substantial toxicity and is contraindicated in older patients; for such patients, less aggressive first-line regimens are recommended, such as bendamustine plus rituximab (BR), acalabrutinib plus BR, R-CHOP (rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone), or VR-CAP (bortezomib, rituximab, cyclophosphamide, doxorubicin, and prednisone).⁶⁻⁸ Based on results from the ENRICH study, in which patients aged ≥ 60 years were randomly assigned to receive ibrutinib plus rituximab or immunochemotherapy, ibrutinib plus rituximab could be considered a new standard-of-care option for the first-line treatment of MCL in this population.⁹

TP53 mutations occur in 15% to 20% of patients with MCL^{10,11} and confer a high risk of early progression and poor outcomes with aggressive chemoimmunotherapy.¹²⁻¹⁴

Ibrutinib, a once-daily oral Bruton tyrosine kinase inhibitor, and venetoclax, a once-daily oral B-cell lymphoma 2 inhibitor, have both shown antitumor activity as single agents in patients with MCL.¹⁵⁻²⁰ The combination of ibrutinib and venetoclax leverages complementary mechanisms of action and has shown

synergistic antitumor activity in preclinical models of MCL,^{21,22} as well as high complete response (CR) rates in early-phase studies in patients with relapsed/refractory MCL.²³⁻²⁵

The phase 3 SYMPATICO study was conducted to evaluate the efficacy and safety of the combination of ibrutinib and venetoclax in 3 cohorts of patients with relapsed/refractory or previously untreated MCL.²⁵⁻²⁷ In the first 2 cohorts, ibrutinib plus venetoclax was evaluated in patients with relapsed/refractory MCL, comprising an open-label safety run-in phase to assess the safety of concurrent initiation of ibrutinib and venetoclax,²⁵ followed by a double-blind randomization phase to evaluate the efficacy and safety of ibrutinib plus venetoclax vs ibrutinib plus placebo.²⁶ Primary analysis results of the randomized phase showed superior PFS with ibrutinib plus venetoclax vs ibrutinib plus placebo (median, 31.9 vs 22.1 months), with a hazard ratio of 0.65 (95% confidence interval [CI], 0.47-0.88; $P = .0052$).²⁶ Additionally, an open-label cohort was enrolled to evaluate the efficacy and safety of first-line ibrutinib plus venetoclax in patients with previously untreated MCL who were aged ≥ 65 years or were aged 18 to < 65 years with *TP53* mutations.²⁷

Here, we report the efficacy and safety of ibrutinib plus venetoclax for previously untreated nonblastoid MCL in patients aged ≥ 65 years with and without *TP53* mutations and in adult patients aged < 65 years with *TP53* mutations.

Methods

Study design and treatment

SYMPATICO is a multicenter, international phase 3 study that included an open-label, single-arm cohort of patients with previously untreated MCL who were aged ≥ 65 years or ≥ 18 years with a *TP53* mutation. Additional eligibility criteria

were pathologically confirmed MCL, at least 1 measurable lesion ≥ 2 cm in diameter and measurable in 2 perpendicular dimensions, Eastern Cooperative Oncology Group performance status score of 0 to 2, and adequate hematologic, hepatic, and renal function. Patients with blastoid variant MCL were excluded.

Patients received oral ibrutinib (560 mg once daily) in combination with oral venetoclax (5-week ramp-up to 400 mg once daily) for 2 years, followed by continuous single-agent ibrutinib (560 mg once daily) until disease progression (PD) or unacceptable toxicity.

The study was conducted in accordance with International Conference on Harmonisation Good Clinical Practice guidelines and provisions of the Declaration of Helsinki. The protocol and amendments were approved by independent ethics committees at each site, and all patients provided written informed consent. This study was registered with [ClinicalTrials.gov](https://clinicaltrials.gov) (identifier: NCT03112174).

Outcomes and assessments

The primary end point was the CR rate as assessed by investigators using Revised Response Criteria for Malignant Lymphoma (Lugano criteria).²⁸ Secondary end points included overall response rate (partial response or CR per Lugano criteria), duration of response and duration of CR (time from first documentation of response until PD or death), minimal residual disease (MRD)-negative remission rate (MCL cells $<0.05\%$ by 8-color flow cytometry, assessed in peripheral blood and bone marrow) in patients who were MRD positive at screening and achieved CR, PFS (time from the first dose of study treatment until PD or death from any cause), overall survival (OS; time from the first dose of study treatment until death from any cause), time to next treatment (time from the first dose of study treatment until initiation of any subsequent antilymphoma therapy), and safety.

Response and progression were assessed clinically and radiographically using computed tomography (with contrast unless contraindicated) of the neck, chest, abdomen, and pelvis and any other disease sites at baseline, every 12 weeks in the first year, every 16 weeks in years 2 and 3, and every 24 weeks thereafter until PD or as clinically indicated. Positron emission tomography scans were obtained at baseline, at week 13, and to confirm a CR. Magnetic resonance imaging could be used for lesions that were not well visualized by computed tomography. Response assessments were based on investigator evaluation without independent central imaging review. Bone marrow biopsies and positron emission tomography scans were mandatory at baseline and were performed as needed thereafter to confirm CR. Treatment-emergent adverse events (TEAEs) were collected during and up to 30 days after the last dose of study treatment and were graded according to Common Terminology Criteria for Adverse Events, version 4.03. Monitoring for tumor lysis syndrome (TLS) was performed using laboratory tests and physical examinations, and TLS was assessed per Howard criteria.²⁹

Somatic mutations in exons 1 to 11 of *TP53* were evaluated by next-generation sequencing, with a variant allele fraction cutoff of 2%; deletions were not assessed.

Statistical analysis

Sample size was calculated based on the primary end point of CR rate. Assuming a target CR rate of 51% with ibrutinib plus venetoclax, a sample size of 75 patients would provide 80% power to exclude a minimum CR rate of 35% at a 1-sided α of 0.025.

Time-to-event end points were estimated using the Kaplan-Meier method. Other efficacy end points and TEAEs were summarized descriptively. Efficacy and safety outcomes were assessed in the all-treated population, comprising all patients who received ≥ 1 dose of study treatment. Prespecified subgroup analyses were performed according to *TP53* mutation status by central laboratory (yes vs no) and age (≥ 65 vs <65 years).

Results

Patients

In total, 78 patients with previously untreated MCL were enrolled and treated with ibrutinib plus venetoclax. The median age was 70 years (range, 41-86). Most patients (83%) were aged ≥ 65 years, and 29 patients had mutated *TP53* (37%); 18 patients (23%) were both aged ≥ 65 years and had mutated *TP53*. Other poor prognostic features were also present in a substantial proportion of patients, including high risk by simplified MCL International Prognostic Index (45%), bulky disease ≥ 5 cm (31%), and extranodal disease (50%) (Table 1).

At data cutoff (31 July 2024), the median time on study was 40.5 months (range, >0.6 -46.9). Treatment with single-agent ibrutinib was ongoing in 26 patients (33%) and had been discontinued in 52 (67%). A total of 39 patients (50%) completed the planned 2 years of treatment with venetoclax, and venetoclax was discontinued early in 39 patients (50%). PD (per protocol criteria) was the most common primary reason for discontinuation of both agents (ibrutinib, $n = 18$ [23%]; venetoclax, $n = 17$ [22%]), followed by adverse events (AEs) not related to PD (ibrutinib, $n = 15$ [19%]; venetoclax, $n = 11$ [14%]).

Efficacy

The primary end point was met, with a CR rate of 69% (95% CI, 58-79) by investigator assessment overall in this cohort (Figure 1A). The overall response rate was 95% (95% CI, 87-99). The median duration of CR was 37.1 months (95% CI, 34.0 to not estimable [NE]; Figure 1B), and the median duration of response was 37.1 months (95% CI, 30.3 to NE; Figure 1C). With a median follow-up duration of 40.5 months, PD or death had occurred in 34 patients. The median PFS was 40.2 months (95% CI, 29.4 to NE), and the 3-year PFS rate was 59% (95% CI, 46-69; Figure 2A). At this final analysis, 20 patients had died. The median OS was not reached, with a 3-year OS rate of 79% (95% CI, 68-86; Figure 2B). Among patients who achieved a CR ($n = 54$) and were evaluable for MRD, MRD-negative remission was achieved in the bone marrow in 13 of 22 evaluable patients (59%) and in the peripheral blood in 26 of 34 evaluable patients (76%). In patients who achieved a MRD-negative remission in the bone marrow ($n = 13$) and peripheral blood ($n = 26$), the 3-year PFS rates were 92% and 79%, respectively; median PFS and OS were not reached in either group (supplemental Table 1, available on the *Blood* website).

Table 1. Baseline demographics and clinical characteristics

Characteristic	All patients (N = 78)
Age, y	
Median (range)	70 (41-86)
≥65, n (%)	65 (83)
Sex, n (%)	
Male	53 (68)
Female	25 (32)
ECOG PS, n (%)	
0	37 (47)
1 or 2	41 (53)
MCL histology, n (%)	
Typical*	56 (72)
Blastoid†	1 (1)
Pleomorphic	6 (8)
Round cell (CLL-like)	2 (3)
Other	13 (17)
Confirmation of diagnosis, n (%)	
t(11;14)	19 (24)
Cyclin D1 with other relevant biomarkers	71 (91)
SOX11 ⁺	8 (10)
CD5 ⁺ /CD20 ⁺	21 (27)
Other	2 (3)
Simplified MIPI score, n (%)	
Low risk	6 (8)
Intermediate risk	37 (47)
High risk	35 (45)
Bulky disease, n (%)	
≥5 cm	24 (31)
≥10 cm	5 (6)
Extranodal disease, n (%)	39 (50)
Bone marrow involvement, n (%)	61 (78)
Splenomegaly, n (%)	36 (46)
TP53 mutation, n (%)‡	29 (37)

CLL, chronic lymphocytic leukemia; ECOG PS, Eastern Cooperative Oncology Group performance status; MIPI, MCL International Prognostic Index.

*Includes 3 patients with "classic" histology as reported by the investigator.

†One patient with blastoid variant MCL was enrolled in error (protocol violation), because this was an exclusion criterion.

‡Among 26 patients with TP53 mutations confirmed by central testing, variant allele frequencies ranged from 11.2% to 99.8%.

The median time to next treatment was not reached, with 66% of patients (95% CI, 53-77) remaining free from next-line therapy at 3 years (Figure 2C). Overall, 19 patients (24%) received subsequent anticancer therapy; of the 14 patients with response data available, all 10 responders (71%) achieved CR (supplemental Table 2).

As expected, outcomes were better in patients without TP53 mutations vs those with TP53 mutations, with CR rates of 77% (95% CI, 62-89) vs 55% (95% CI, 36-74; Figure 3A; Table 2). In addition, the median PFS was 40.2 months (95% CI, 37.2 to NE) vs 22.0 months (95% CI, 9.2 to NE; Figure 3B), and the 3-year OS rates were 86% (95% CI, 71-93) vs 68% (95% CI, 47-82; Figure 3C). Worse outcomes in patients with TP53 mutations are not explained by modest differences in baseline characteristics between patients with vs without TP53 mutations (supplemental Table 3) and are most likely attributable to the presence of TP53 mutations. Subgroup analyses examining the outcomes by age and TP53 mutation status were also performed. In patients aged ≥65 years without TP53 mutations, the CR rate was 76% (95% CI, 61-88). In contrast, the CR rate was 44% (95% CI, 22-69) in patients aged ≥65 years with TP53 mutations and 73% (95% CI, 39-94) in those aged <65 years with TP53 mutations (supplemental Figure 1A; supplemental Table 4). The median PFS was 40.2 months (95% CI, 37.2 to NE) in patients aged ≥65 years without TP53 mutations, 22.0 months (95% CI, 11.3 to NE) in those aged ≥65 years with TP53 mutations, and 15.4 months (95% CI, 8.2 to NE) in those aged <65 years with TP53 mutations (supplemental Figure 1B-C); 3-year OS rates were 85% (95% CI, 70-93), 66% (95% CI, 39-83), and 73% (95% CI, 37-90), respectively (supplemental Figure 1D-E).

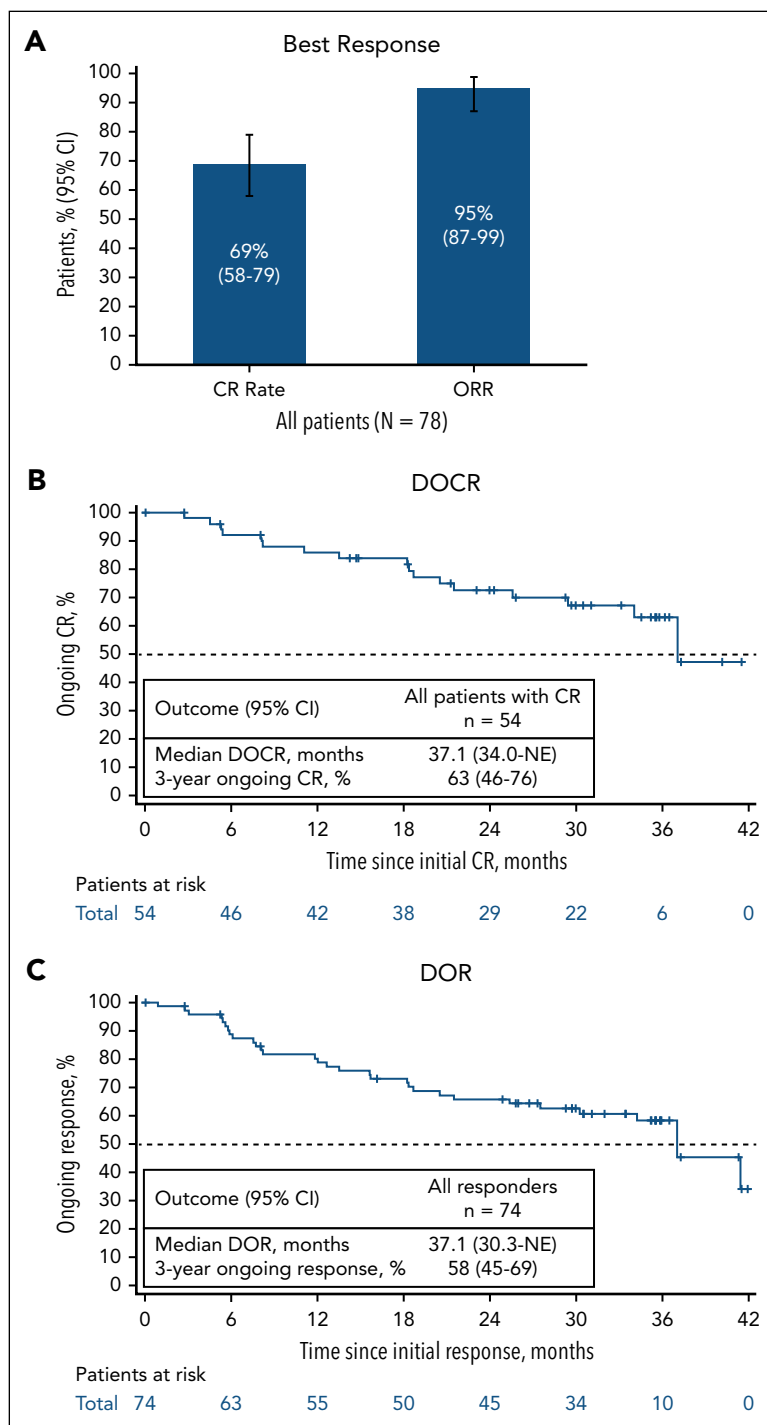
Safety

The overall median duration of treatment was 24.0 months (range, 0.3-46.9). The median duration of treatment with ibrutinib was 21.9 months (range, 0.3-46.9), and the median duration of treatment with venetoclax was 22.9 months (range, 0.3-25.3). All patients experienced at least 1 TEAE (Table 3). The most common TEAEs of any grade (occurring in ≥30% of patients) were diarrhea (49%), fatigue (37%), neutropenia (35%), and COVID-19 (32%). Grade ≥3 TEAEs occurred in 67 patients (86%); the most common (occurring in ≥10% of patients) were neutropenia (29%), hypertension (12%), and neutrophil count decreased (10%). Serious TEAEs occurred in 46 patients (59%); the most common (occurring in ≥5% of patients) were COVID-19 (8%), COVID-19 pneumonia (8%), and pneumonia (6%). TEAEs leading to death occurred in 7 patients (9%), all of whom were aged ≥65 years. TEAEs leading to death were COVID-19 (n = 2), pneumonia (n = 1), MCL (n = 1), acute respiratory failure (n = 1), cardiac arrest (n = 1), and cardiac failure (n = 1). The incidence of grade ≥3 TEAEs was numerically higher in patients aged ≥65 vs <65 years (88% vs 77%), as was the incidence of serious TEAEs (62% vs 46%). The safety profile in patients with TP53 mutations was consistent with that observed in the overall population (supplemental Table 5).

AEs of special interest included atrial fibrillation, which occurred in 13 patients (17%), with grade 3 or 4 events occurring in 4 patients (5%); no grade 5 atrial fibrillation events occurred. Laboratory TLS was reported as a TEAE in 5 patients (6%), all of whom were aged ≥65 years. No clinical TLS was observed.

TEAEs led to dose reductions of ibrutinib and/or venetoclax in 34 patients (44%) and discontinuation of ibrutinib and/or venetoclax in 24 patients (31%; Table 3). TEAEs led to

Figure 1. Best response and DOR. Bar graphs of best response per investigator assessment (A), Kaplan-Meier curves of DOCR in patients who achieved a CR (B), and DOR in all responders (C). DOCR, duration of CR; DOR, duration of response; ORR, overall response rate.



discontinuation of ibrutinib in 22 patients (28%) and discontinuation of venetoclax in 17 patients (22%; [Table 3](#)). The most common TEAEs leading to discontinuation of ibrutinib (occurring in $\geq 2\%$ of patients) were atrial fibrillation (5%), diarrhea (4%), and COVID-19 infection (3%); the most common TEAEs leading to discontinuation of venetoclax were COVID-19 infection (3%), diarrhea (3%), and fatigue (3%).

TEAEs leading to discontinuation of ibrutinib and/or venetoclax were higher in patients aged ≥ 65 vs < 65 years (34% vs 15%; [Table 3](#)). TEAEs leading to dose reductions of ibrutinib

only were numerically higher in patients aged ≥ 65 vs < 65 years (23% vs 15%), as were TEAEs leading to dose reductions of both ibrutinib and venetoclax (17% vs 8%), whereas TEAEs leading to dose reductions of venetoclax only were lower in patients aged ≥ 65 vs < 65 years (2% vs 31%; [Table 3](#)).

Discussion

Unmet needs persist for patients with MCL, especially those with *TP53* mutations, due to high risk of early PD and poor outcomes with chemoimmunotherapy,^{5,9,12-14,30-35} and for

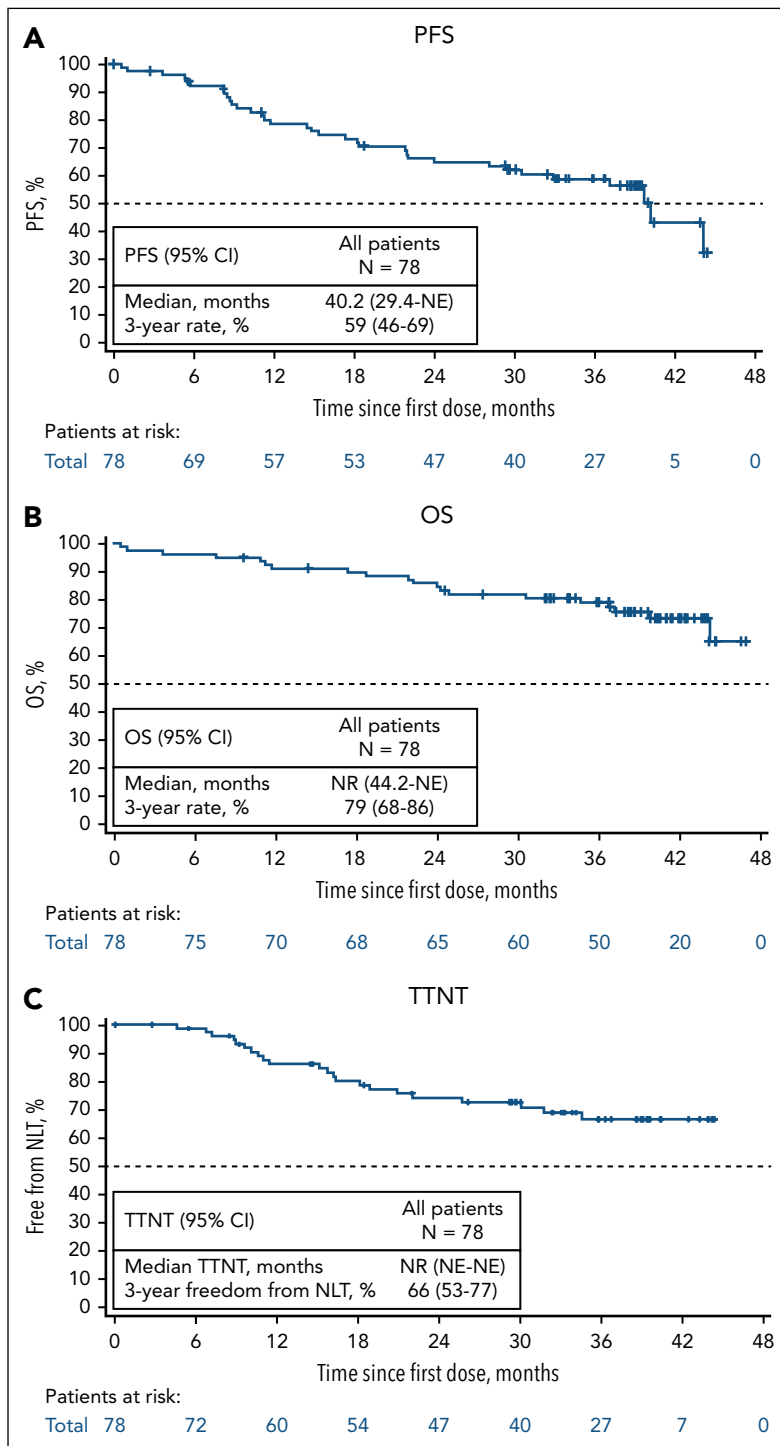


Figure 2. PFS, OS, and TTNT. Kaplan-Meier curves of PFS per investigator assessment (A), OS (B), and TTNT (C). For PFS, patients without progression or those who had not died were censored at the last adequate disease assessment. For OS, patients who had not died were censored at the last date known to be alive. For TTNT, patients without subsequent anticancer treatment were censored at the last follow-up date. Tick marks indicate censored patients. NLT, next-line therapy; NR, not reached; TTNT, time to next treatment.

older patients due to an intolerance of the most effective intensive chemoimmunotherapy.^{2,3} Results from this study show that first-line ibrutinib plus venetoclax had promising efficacy, with high CR rates and durable remissions in patients with nonblastoid MCL, including both younger and older patients and those with and without *TP53* mutations. Although it is acknowledged that ibrutinib is not approved for the treatment of MCL in the United States, ibrutinib remains available in many other countries, and the combination of

ibrutinib and venetoclax has an important role in the treatment of MCL globally. The ibrutinib plus venetoclax combination is currently approved for the treatment of relapsed/refractory MCL in several countries.

For previously untreated MCL, the CR rate of 69% observed with ibrutinib plus venetoclax was promising given the age and prevalence of *TP53* mutations in the overall population of 78 patients, including 83% (n = 65) who were aged ≥ 65 years and

Figure 3. Response rates, PFS, and OS in subgroups by TP53 mutation status. Bar graphs of CR rates and ORR by TP53 mutation status (A), Kaplan-Meier curves of PFS by TP53 mutation status (B), and OS by TP53 mutation status (C). For PFS, patients without progression or those who had not died were censored at the last adequate disease assessment. For OS, patients who had not died were censored at the last date known to be alive. Tick marks indicate censored patients. NR, not reached; ORR, overall response rate.

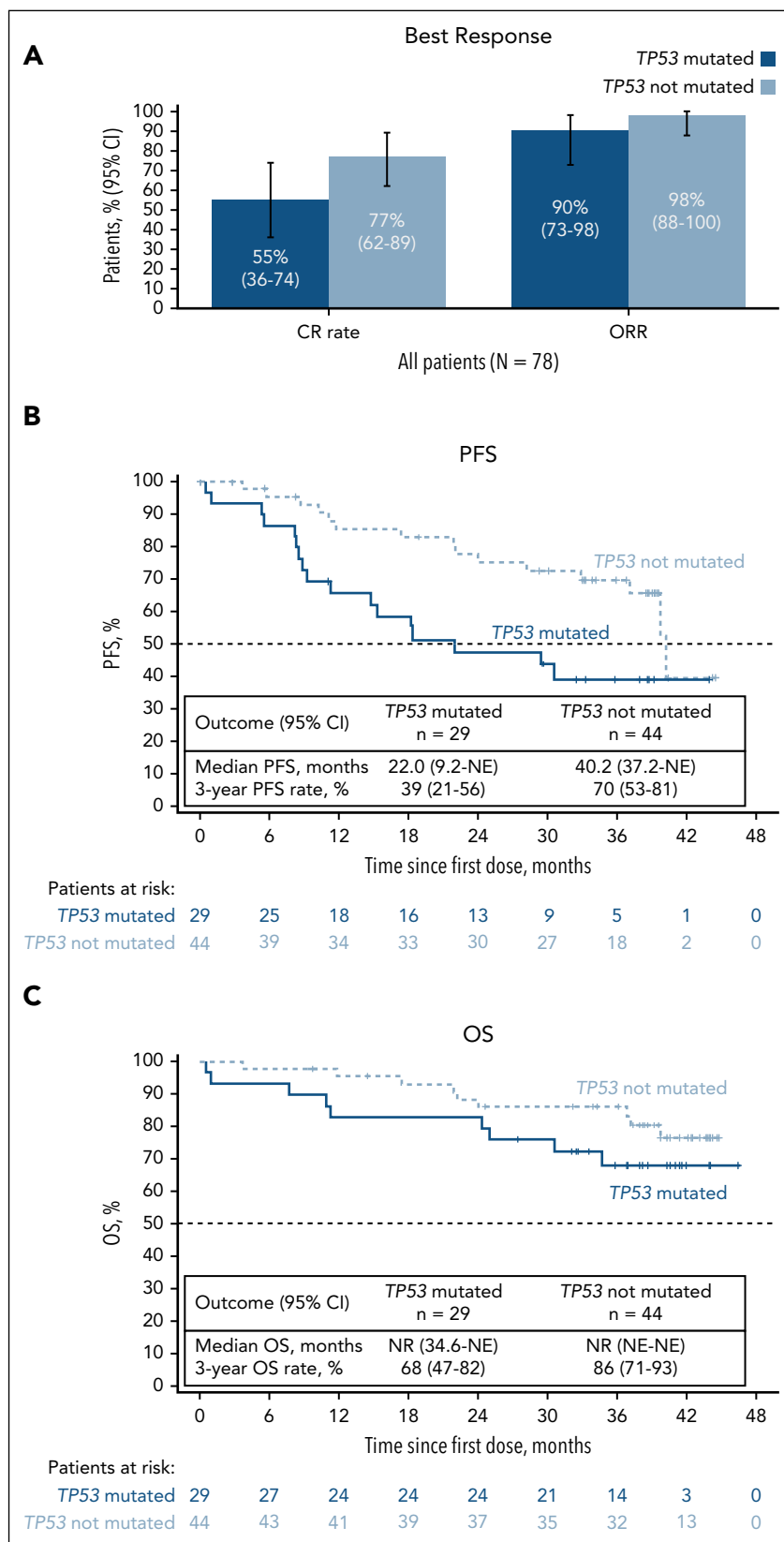


Table 2. Response and duration of response by TP53 mutation status

Outcome	All patients	
	TP53 mutated (n = 29)	TP53 not mutated (n = 44)
CR rate (95% CI), %	55 (36-74)	77 (62-89)
DOCR, median (95% CI), mo	NR (5.4 to NE)	NR (34.0 to NE)
ORR (95% CI), %	90 (73-98)	98 (88-100)
DOR, median (95% CI), mo	20.5 (12.0 to NE)	37.1 (34.2 to NE)

DOCR, duration of CR; DOR, duration of response; NE, not estimable; NR, not reached; ORR, overall response rate.

Table 3. TEAEs

AEs, n (%)	All patients (N = 78)	Patients aged ≥65 y (n = 65)	Patients aged <65 y (n = 13)
AEs of any grade	78 (100)	65 (100)	13 (100)
Grade ≥3 AEs*	67 (86)	57 (88)	10 (77)
Grade 1-2 AEs*	11 (14)	8 (12)	3 (23)
Serious AEs	46 (59)	40 (62)	6 (46)
AEs leading to discontinuation	24 (31)	22 (34)	2 (15)
Ibrutinib only	7 (9)	7 (11)	0
Venetoclax only	2 (3)	2 (3)	0
Both	15 (19)	13 (20)	2 (15)
AEs leading to dose reduction	34 (44)	27 (42)	7 (54)
Ibrutinib only	17 (22)	15 (23)	2 (15)
Venetoclax only	5 (6)	1 (2)	4 (31)
Both	12 (15)	11 (17)	1 (8)
AEs leading to death	7 (9)	7 (11)	0
Most frequent any-grade AEs†			
Diarrhea	38 (49)	31 (48)	7 (54)
Fatigue	29 (37)	26 (40)	3 (23)
Neutropenia	27 (35)	23 (35)	4 (31)
COVID-19	25 (32)	22 (34)	3 (23)
Nausea	23 (29)	17 (26)	6 (46)
Pyrexia	19 (24)	15 (23)	4 (31)
Anemia	17 (22)	14 (22)	3 (23)
Dizziness	17 (22)	15 (23)	2 (15)
Increased tendency to bruise	17 (22)	16 (25)	1 (8)
Dyspnea	16 (21)	13 (20)	3 (23)
Hypomagnesemia	16 (21)	12 (18)	4 (31)
Hypertension	16 (21)	13 (20)	3 (23)
Myalgia	16 (21)	13 (20)	3 (23)
Vomiting	16 (21)	11 (17)	5 (38)

*Maximum reported grade.

†Occurring in ≥20% of patients overall.

‡Occurring in ≥5% of patients overall.

Table 3 (continued)

AEs, n (%)	All patients (N = 78)	Patients aged ≥65 y (n = 65)	Patients aged <65 y (n = 13)
Most frequent grade ≥3 AEs*,†			
Neutropenia	23 (29)	19 (29)	4 (31)
Hypertension	9 (12)	7 (11)	2 (15)
Neutrophil count decreased	8 (10)	6 (9)	2 (15)
COVID-19	7 (9)	7 (11)	0
Diarrhea	7 (9)	5 (8)	2 (15)
COVID-19 pneumonia	6 (8)	6 (9)	0
Fatigue	6 (8)	5 (8)	1 (8)
Rash maculopapular	5 (6)	5 (8)	0
Anemia	4 (5)	3 (5)	1 (8)
Atrial fibrillation	4 (5)	3 (5)	1 (8)
Pneumonia	4 (5)	3 (5)	1 (8)
TLS	4 (5)	4 (6)	0
Most frequent grade 1-2 AEs*,‡			
Diarrhea	31 (40)	26 (40)	5 (38)
Fatigue	23 (29)	21 (32)	2 (15)
Nausea	22 (28)	16 (25)	6 (46)
COVID-19	18 (23)	15 (23)	3 (23)
Pyrexia	17 (22)	14 (22)	3 (23)
Dizziness	17 (22)	15 (23)	2 (15)
Increased tendency to bruise	17 (22)	16 (25)	1 (8)
Myalgia	16 (21)	13 (20)	3 (23)

*Maximum reported grade.

†Occurring in ≥20% of patients overall.

‡Occurring in ≥5% of patients overall.

37% (n = 29) who had *TP53* mutations regardless of age. Although cross-trial comparisons should be interpreted with caution due to differences in patient populations, these rates appear favorable in the context of historical CR rates of 40% to 50% observed with rituximab-based chemoimmunotherapy regimens in patients aged ≥65 years.³⁶⁻³⁸ CRs were durable, with a median duration of 37.1 months. The median PFS was 40.2 months in patients aged ≥65 years without *TP53* mutations; as expected, the median PFS was lower in patients with *TP53* mutations (22 months in patients aged ≥65 years and 15.4 months in those aged <65 years) but favorable compared with other reports.^{13,39} The overall 3-year PFS and OS rates of 59% and 79%, respectively, also appear favorable relative to rituximab-based chemoimmunotherapy,^{36-38,40} with particularly high 3-year PFS rates in patients who achieved MRD-negative remission in the peripheral blood (79%) or bone marrow (92%). The CR rate achieved with ibrutinib plus venetoclax in this study (69%) appears favorable relative to a CR rate of 54% observed with ibrutinib plus rituximab in patients aged ≥60 years in the ENRICH study⁹ and was commensurate with CR rates reported in older patients who were treated with ibrutinib plus BR in the phase 3 SHINE study (66% in 261 patients, 10% of whom had *TP53* mutations)⁴¹ or with acalabrutinib plus BR in the phase 3 ECHO study (67% in 299 patients, 7% of whom had *TP53* mutations).⁴²

No standard first-line treatment approach has been defined for patients with MCL harboring *TP53* mutations, and clinical trial

participation is recommended for these patients.⁸ The CR rate of 55% among patients with *TP53* mutations treated with ibrutinib plus venetoclax in this study appears promising relative to a reported CR rate of 25% achieved with induction chemoimmunotherapy.^{13,39} Additionally, as mentioned above, the median PFS of 22.0 months in patients with *TP53* mutations also appears promising compared with the median PFS durations of <1 year with conventional chemoimmunotherapy regimens.^{13,39} Interestingly, although ibrutinib plus venetoclax was better tolerated in younger patients, PFS and OS were similar in patients with *TP53* mutations regardless of age in these small subgroups. Promising response rates and PFS were also recently reported in 25 patients with *TP53* mutations who received first-line treatment with the triplet combination of zanubrutinib plus obinutuzumab plus venetoclax in the BoVEN study, with a CR rate of 88% and a 2-year PFS rate of 72%.⁴³

A limitation of the efficacy analysis may be that response assessments were based on investigator evaluation without independent central imaging review. Additionally, the use of flow cytometry for MRD testing may have failed to detect some cases of MRD compared with other more sensitive methods. Interpretation of outcomes for younger patients (aged <65 years) with a *TP53* mutation is limited by the small number of patients (n = 11) in this subgroup. Lastly, exclusion of patients with blastoid histology may limit the generalizability of results for such patients.

The safety profile of ibrutinib plus venetoclax in patients with previously untreated MCL was consistent with known safety profiles of the individual agents, with no new safety signals observed. The incidence of grade ≥ 3 AEs was higher in patients aged ≥ 65 years than in those aged < 65 years (88% vs 77%), as was the incidence of AEs leading to dose reductions of both study drugs (17% vs 8%) and AEs leading to discontinuation of either or both study drugs (34% vs 15%). These results are consistent with data from previous studies suggesting that rates of AEs leading to discontinuation of ibrutinib are higher in older patients who have a greater comorbidity burden overall, which may have affected the efficacy in these patients.^{44,45} The incidence of atrial fibrillation was consistent with incidence rates previously reported in older adults treated with ibrutinib.^{44,45}

In conclusion, first-line ibrutinib plus venetoclax showed promising efficacy, with high CR rates and durable remissions and a manageable safety profile, in patients with previously untreated, nonblastoid MCL and may be an option for older patients or patients of any age with *TP53* mutations.

Acknowledgments

AbbVie and the authors thank the patients who participated in the study and their supportive families, as well as the investigators, study coordinators, study teams, and nurses who cared for the patients. This study was funded by AbbVie. Medical writing support was provided by Melanie Sweetlove and funded by AbbVie.

AbbVie participated in the study design, research, analysis, data collection, interpretation of data, and the review and approval of the publication.

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Conflict-of-interest disclosure: M.W. reports honoraria from Acerta Pharma, Anticancer Association, AstraZeneca, BeiGene, BGICS, Bio-Invent, Chinese American Hematologist and Oncologist Network, Chinese Medical Association, Clinical Care Options, Dava Oncology, Eastern Virginia Medical School, Epizyme, Hebei Cancer Prevention Federation, Imedex, Janssen, Kite Pharma LLC, Miltenyi Biomedicine GmbH, Moffitt Cancer Center, Mumbai Hematology Group, Newbridge Pharmaceuticals, OMI, Onclive, Physicians Education Resources (PER), Practice Point Communications, Scripps, The First Affiliated Hospital of Zhejiang University, TS Oncology, and Pharmacyclics LLC, an AbbVie company; consulting/advisory role for AstraZeneca, Bayer Healthcare, BeiGene, BioInvent, CStone Pharmaceuticals, DTRM Biopharma (Cayman) Limited, Epizyme, Genentech, InnoCare, Janssen, Juno Therapeutics, Kite Pharma, Loxo Oncology, Miltenyi Biomedicine GmbH, Oncernal, VelosBio, and Pharmacyclics LLC, an AbbVie company; research funding from Acerta Pharma, AstraZeneca, BeiGene, BioInvent, Celgene, Genentech, InnoCare, Janssen, Juno Therapeutics, Kite Pharma, Lilly, Loxo Oncology, Molecular Templates, Oncernal, VelosBio, and Pharmacyclics LLC, an AbbVie company; and travel/accommodations/expenses from Acerta Pharma, AstraZeneca, Bio-Invent, Celgene, Janssen, Juno Therapeutics, Kite, Loxo Oncology, Molecular Templates, Oncernal, PER, VelosBio, Verastem, and Pharmacyclics LLC, an AbbVie company. M.H. reports honoraria from and consulting/advisory role for AstraZeneca, BeiGene, Janssen, Kite, Novartis, TG Therapeutics, and Pharmacyclics LLC, an AbbVie company. T.W. reports honoraria from BeiGene, Bristol Myers Squibb, Celgene, Gilead, Janssen-Cilag, Novartis, Roche, Sanofi, and Takeda; consulting/advisory role for BeiGene, Bristol Myers Squibb, Celgene,

Gilead, GSK, Janssen-Cilag, Novartis, Pfizer, Roche, Sanofi, and Takeda; research funding from Roche; speakers bureau fees from BeiGene, Bristol Myers Squibb, Celgene, Janssen-Cilag, Novartis, Roche, Sanofi, and Takeda; and travel/accommodations/expenses from Gilead, Roche, and Takeda. M.T. reports honoraria from AbbVie, Amgen, Bristol Myers Squibb, Gilead Sciences, Janssen, MorphoSys, Novartis, Roche, Sobi, Swix BioPharma, and Takeda; consulting/advisory role for AbbVie, Amgen, Autolus, Bristol Myers Squibb, Caribou Biosciences, Genmab, Gilead, Incyte, Janssen, MorphoSys, Novartis, Roche, Sobi, and Takeda; and travel/accommodations/expenses from AbbVie, Bristol Myers Squibb, Gilead, Janssen, Roche, Sobi, and Takeda. D.B. reports a consulting/advisory role for Gilead, Janssen-Cilag, Novartis, Roche, and Takeda; research funding from Genmab, Janssen-Cilag, MorphoSys, and Roche and travel/accommodations/expenses from Gilead, Roche, and Takeda. P.P. reports honoraria from AbbVie, Genesis Pharma, Gilead, Janssen, and Novartis and research funding from AbbVie, Genesis Pharma, Novartis, and Roche. W.J. reports a consulting/advisory role for AstraZeneca, BeiGene, Janssen, Loxo Oncology, Roche, and Sandoz and research funding from AbbVie, AstraZeneca, Bayer, BeiGene, Celltrion, Celgene, DebioPharm, Epizyme, Incyte, Janssen, Loxo Oncology, MEI Pharma, Merck, MorphoSys, Novo Nordisk, Roche, Sandoz, Takeda, and TG Therapeutics. P.L.Z. reports a consulting/advisory role for ADC Therapeutics, AstraZeneca, BeiGene, Bristol Myers Squibb, Incyte, Janssen, Kite-Gilead, Kyowa Kirin, Merck, Novartis, Recordati, Roche, Secura Bio, Servier, Sobi, and Takeda and speakers bureau fees from AstraZeneca, BeiGene, Bristol Myers Squibb, Incyte, Janssen, Kite-Gilead, Kyowa Kirin, Merck, Novartis, Recordati, Roche, Servier, Sobi, and Takeda. M.-M.K. reports a consulting/advisory role for AbbVie, AstraZeneca, BeiGene, Eli Lilly, and Roche. S.-S.Y. reports honoraria from Celgene, Janssen, and Novartis; consulting/advisory role for Amgen, Antengene, Janssen, Novartis, Regeneron, Sanofi, and Takeda; and research funding from Chong Kun Dang Pharmaceutical Corp, JW Pharmaceutical Corporation, Kyowa Kirin, Roche/Genentech, and Yuhan. C.S.T. reports honoraria from AbbVie, BeiGene, Janssen, and Loxo Oncology and research funding from AbbVie, BeiGene, and Janssen. N.A.J. reports a consulting/advisory role for AbbVie, AstraZeneca, BeiGene, Gilead, Incyte, Lilly, and Roche/Genentech; honoraria from AbbVie, AstraZeneca, and Incyte; and research funding from Gilead, Incyte, and Roche. E.S.-G., J.L., J.P.D., and J.K.N. report employment with and stock ownership in AbbVie. G.v.K. reports honoraria from and consulting/advisory role for AbbVie, Incyte, Merck, and Pharmacyclics LLC, an AbbVie company and research funding from AbbVie, Janssen, Merck, Syndax, and TG Therapeutics. The remaining authors declare no competing financial interests.

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Footnotes

Submitted 19 December 2025; accepted 2 June 2026; prepublished online on *Blood* First Edition 1 July 2026. <https://doi.org/10.1182/blood.2025032833>.

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